Tutorial Questions | Week 8

COSC2779 - Deep Learning

This tutorial is aimed at reviewing the practical methodology of applying Transformers to natural language processing. Please try the questions before you join the session. The question below has a similar structure to the case study questions that you can expect in the end semester test.

1. You have been hired by a company that provides customer-experience analytics to a large Australian telecommunications provider. Every month the provider receives millions of free-text support messages from customers, and the company wants to process them automatically.

   They have two immediate needs:

   - **Intent routing**: classify each message with a single label (e.g. *billing*, *technical-fault*, *cancellation*) so it can be sent to the right team.
   - **Entity highlighting**: mark every mention of a product, a location, or an account identifier inside the message (a label for every word).

   They have collected 200,000 messages, each annotated with an intent label and with word-level entity tags. Messages range from a few words to several hundred. You decide to build the system with Transformers.

   (a) The model needs numbers, not raw text. Explain how you would turn a message into the sequence of inputs a Transformer consumes. Why not simply split on whitespace and one-hot encode each word?

   (b) You choose a Transformer over a recurrent network (RNN/LSTM). Give 2 reasons.

   (c) Self-attention is built from Queries, Keys and Values. Briefly say what each is, write the scaled dot-product attention, and explain why we divide by $\sqrt{d_k}$.

   (d) Why use multi-head attention instead of a single attention?

   (e) Self-attention is a set operation – permute the input tokens and the outputs just permute too. Why is that a problem here, and how is it fixed?

   (f) From the same encoder output, how do you produce (i) the single intent label for the message, and (ii) the per-word entity tags?

   (g) You will start from a pretrained model rather than training from scratch. Would you choose an encoder-only (BERT-style), a decoder-only (GPT-style), or an encoder–decoder model for these two tasks? Why, and how do you use it?

   (h) Messages have very different lengths, but you train in batches. What must the attention handle, and how?

   (i) The client now also wants the system to draft a suggested reply to each message. What architecture would you use, and how does the attention change compared to the classifier?

   (j) While fine-tuning the classifier you observe that the **training loss keeps decreasing but the validation loss starts rising after only one or two epochs**. What are the possible alterations you might try?

   (k) Some transcripts are thousands of tokens long, and training runs out of memory on these. Why does this happen, and give one way to mitigate it.

   (l) (Vision aside.) The same company also wants to classify small product screenshots. How can the Transformer encoder be applied to an image?